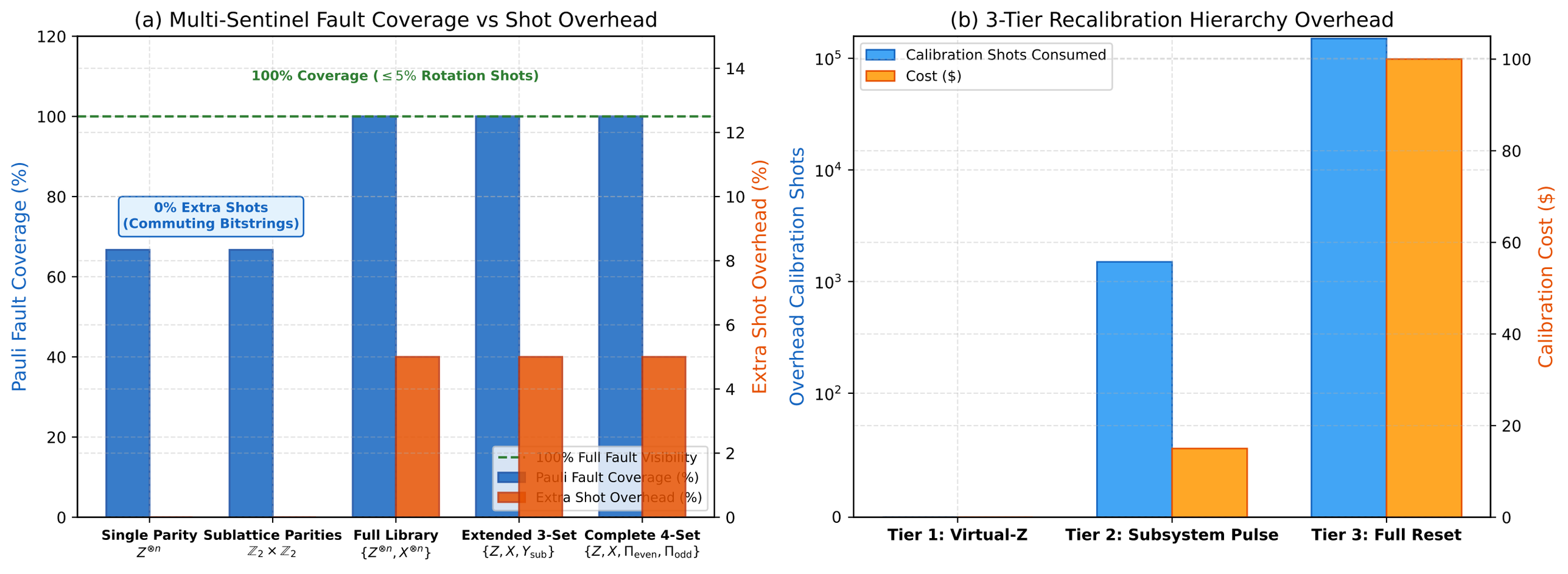
(a) Multi-Sentinel Fault Coverage vs Shot Overhead
Pauli Fault Coverage (%)
Extra Shot Overhead (%)
100% Coverage (≤ 5% Rotation Shots)
0% Extra Shots
(Commuting Bitstrings)
100% Full Fault Visibility
Pauli Fault Coverage (%)
Extra Shot Overhead (%)
Single Parity
$Z^{\otimes n}$
Sublattice Parities
$\mathbb{Z}_2 \times \mathbb{Z}_2$
Full Library
$\{Z^{\otimes n}, X^{\otimes n}\}$
Extended 3-Set
$\{Z, X, Y_{\text{sub}}\}$
Complete 4-Set
$\{Z, X, \Pi_{\text{even}}, \Pi_{\text{odd}}\}$
(b) 3-Tier Recalibration Hierarchy Overhead
Overhead Calibration Shots
Calibration Cost ($)
Calibration Shots Consumed
Cost ($)
Tier 1: Virtual-Z
Tier 2: Subsystem Pulse
Tier 3: Full Reset